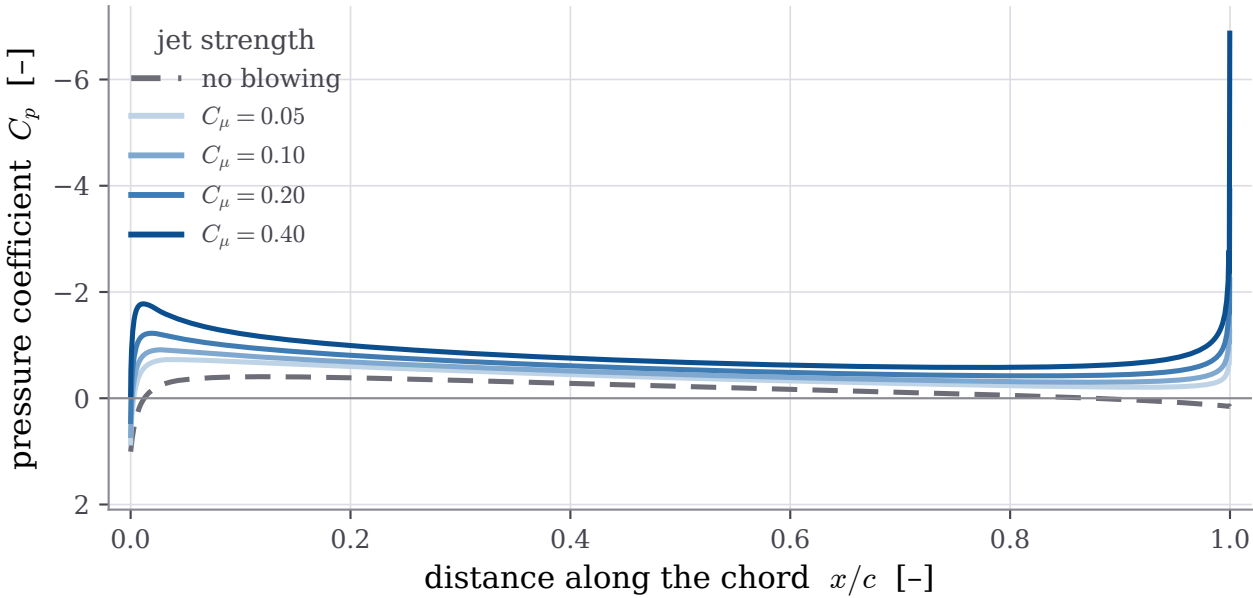


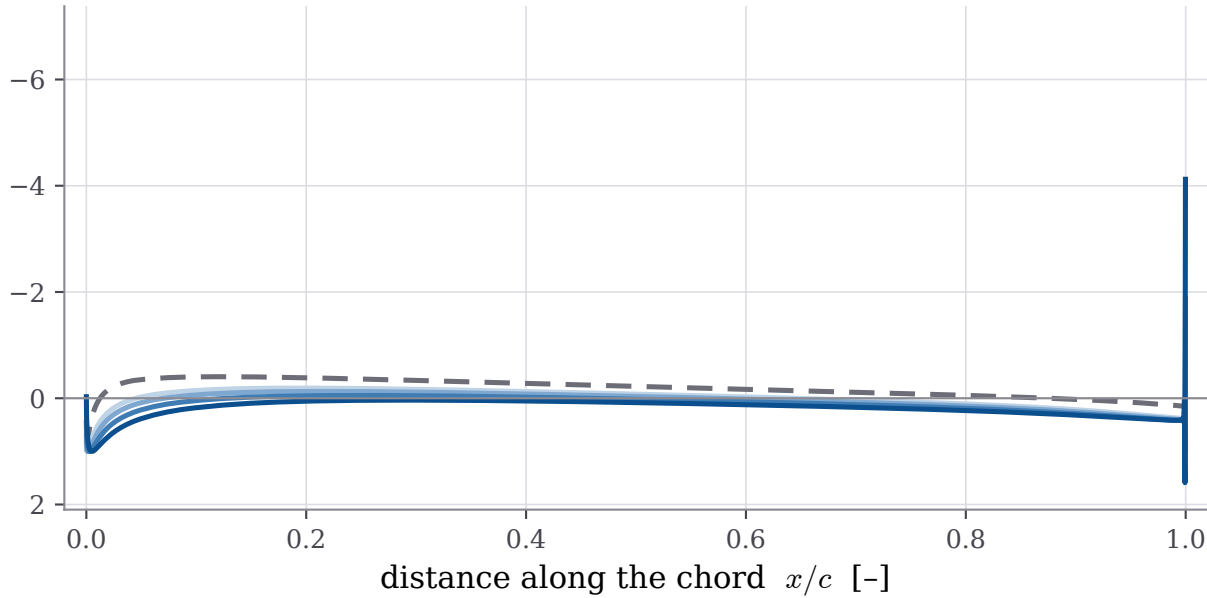
PRELIMINARY — EXPLORATORY CALCULATION, NOT A VALIDATED RESULT

Pressure around the wing, drawn to full scale with nothing cut off the axis. Same wing, same mesh, same flow speed in all five curves; only the strength of the trailing-edge jet changes, so this is a controlled comparison. Suction is plotted upward, the usual aerodynamic convention. The area enclosed between the upper and lower curves is the lift. The lower row magnifies the first 7.5 % of the chord, where every curve reaches  $C_p = +1$  — the point at which the oncoming air is brought to rest. With no jet that point sits on the nose; the stronger the jet, the further back along the lower surface it moves.

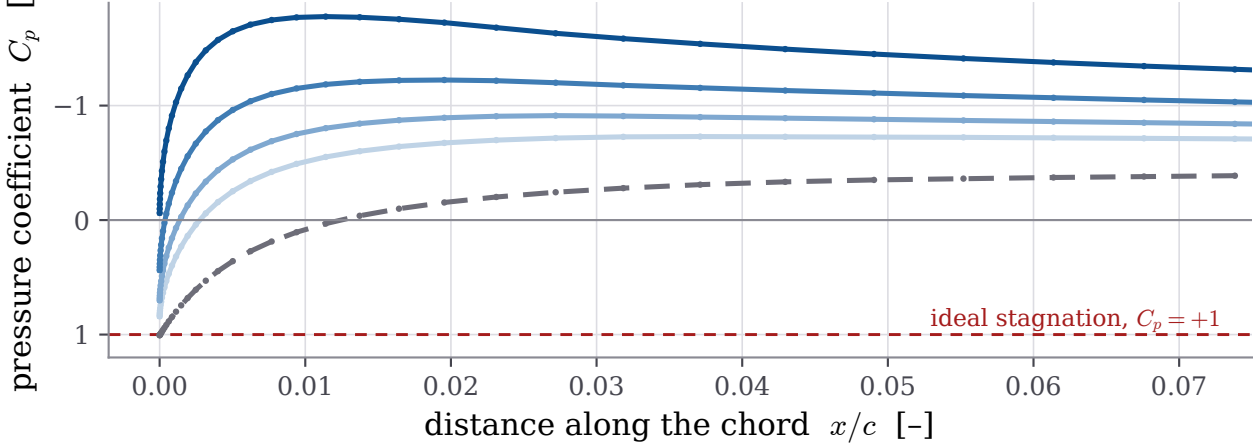
Upper (suction) surface — full scale, nothing cut off



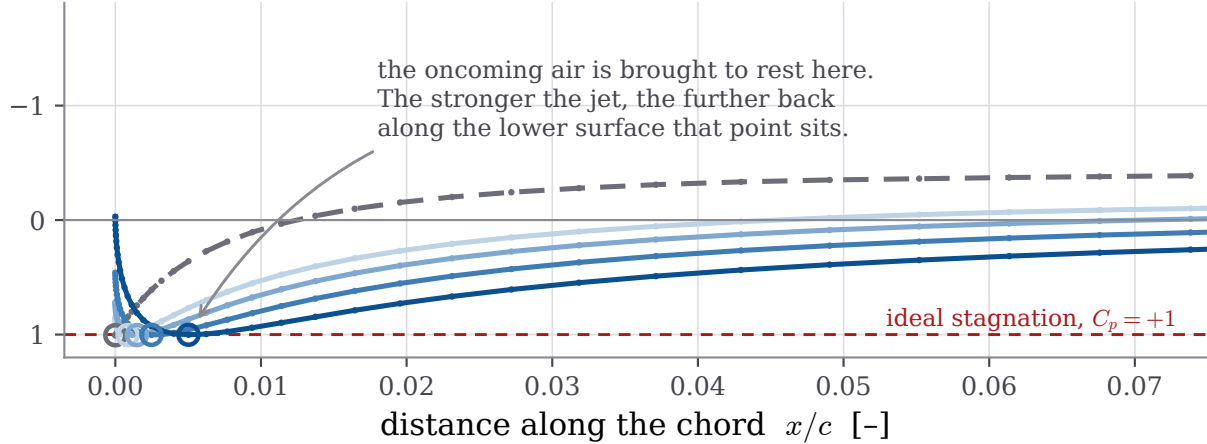
Lower (pressure) surface — full scale, nothing cut off



Leading edge, upper surface (detail)



Leading edge, lower surface (detail)



| MEASURED VALUES BEHIND THIS CHART (all quantities dimensionless; 396 surface samples per curve) |                                  |                           |                             |                     |                                                |                 |                                    |
|-------------------------------------------------------------------------------------------------|----------------------------------|---------------------------|-----------------------------|---------------------|------------------------------------------------|-----------------|------------------------------------|
| jet strength<br>$C_\mu$ [-]                                                                     | stagnation<br>pressure $C_p$ [-] | its location<br>$x/c$ [-] | location<br>uncertainty [-] | on which<br>surface | strongest suction,<br>$x/c < 0.98$ : $C_p$ [-] | at<br>$x/c$ [-] | lowest $C_p$ on<br>the surface [-] |
| none                                                                                            | +1.0057                          | 0.00000                   | $\pm 0.00000$               | lower               | -0.406                                         | 0.118           | -0.406                             |
| 0.05                                                                                            | +1.0049                          | 0.00085                   | $\pm 0.00014$               | lower               | -0.729                                         | 0.037           | -1.258                             |
| 0.10                                                                                            | +1.0046                          | 0.00148                   | $\pm 0.00022$               | lower               | -0.912                                         | 0.027           | -2.236                             |
| 0.20                                                                                            | +1.0027                          | 0.00248                   | $\pm 0.00034$               | lower               | -1.223                                         | 0.020           | -3.916                             |
| 0.40                                                                                            | +1.0015                          | 0.00502                   | $\pm 0.00061$               | lower               | -1.777                                         | 0.011           | -6.873                             |

WHAT YOU SHOULD KNOW ABOUT THESE NUMBERS

- The stagnation pressure comes out between +1.0015 and +1.0057 rather than exactly +1. The overshoot is 0.15 % to 0.57 % and is a resolution effect: pressure is sampled at the centre of each surface cell, not at the exact stagnation point.
- The strong peaks at the trailing edge are the slot lip. They are real features of the calculation and they set the scale of the top row. Nothing is cut off, which is why the mid-chord detail looks flatter here than on a chart with a cropped axis.
- None of these calculations reached the convergence target that was fixed before they were run (all five solution channels below  $1 \times 10^{-6}$ ). The turbulence-energy channel is the slowest everywhere and worsens with blowing —  $3.5 \times 10^{-6}$  with no jet,  $1.5 \times 10^{-4}$  at the strongest jet, a factor of 43.
- Mesh sensitivity has NOT been quantified: no refinement study was run, so no uncertainty is claimed on any pressure value. The one uncertainty in the table is on WHERE the stagnation point sits, and it is half the local surface-cell spacing.
- No experimental pressure data exists for this configuration, so no measured reference curve is drawn. Treat these as indicative.